

Ella Menon

Embedded Systems & IoT Firmware Engineer • 5 Years Experience • MERIQ Index: 72% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Embedded Linux, C/C++, ARM Cortex, FreeRTOS, PCB Debugging, Python, MQTT

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Embedded Linux service, reducing latency by 33%.
- Led a squad of 5 engineers delivering scalable microservices processing 23M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 62%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2021 - 2023)

- Designed and maintained C/C++ data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using ARM Cortex saving \$19k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Embedded Linux, C/C++, ARM Cortex, FreeRTOS]

Engineered end-to-end production solution leveraging Embedded Linux and C/C++ with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [ARM Cortex, FreeRTOS, PCB Debugging, Python]

Architected high-resilience system integrating ARM Cortex with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

University of Toronto — 2017 - 2021 (Score: 7.88 / 10.0)